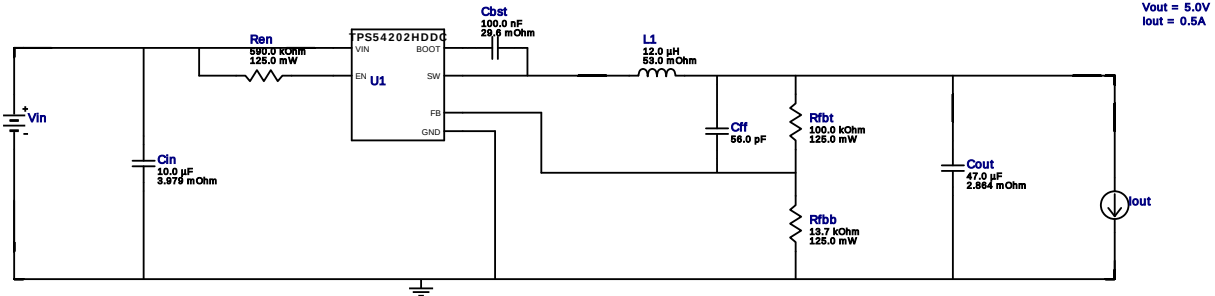


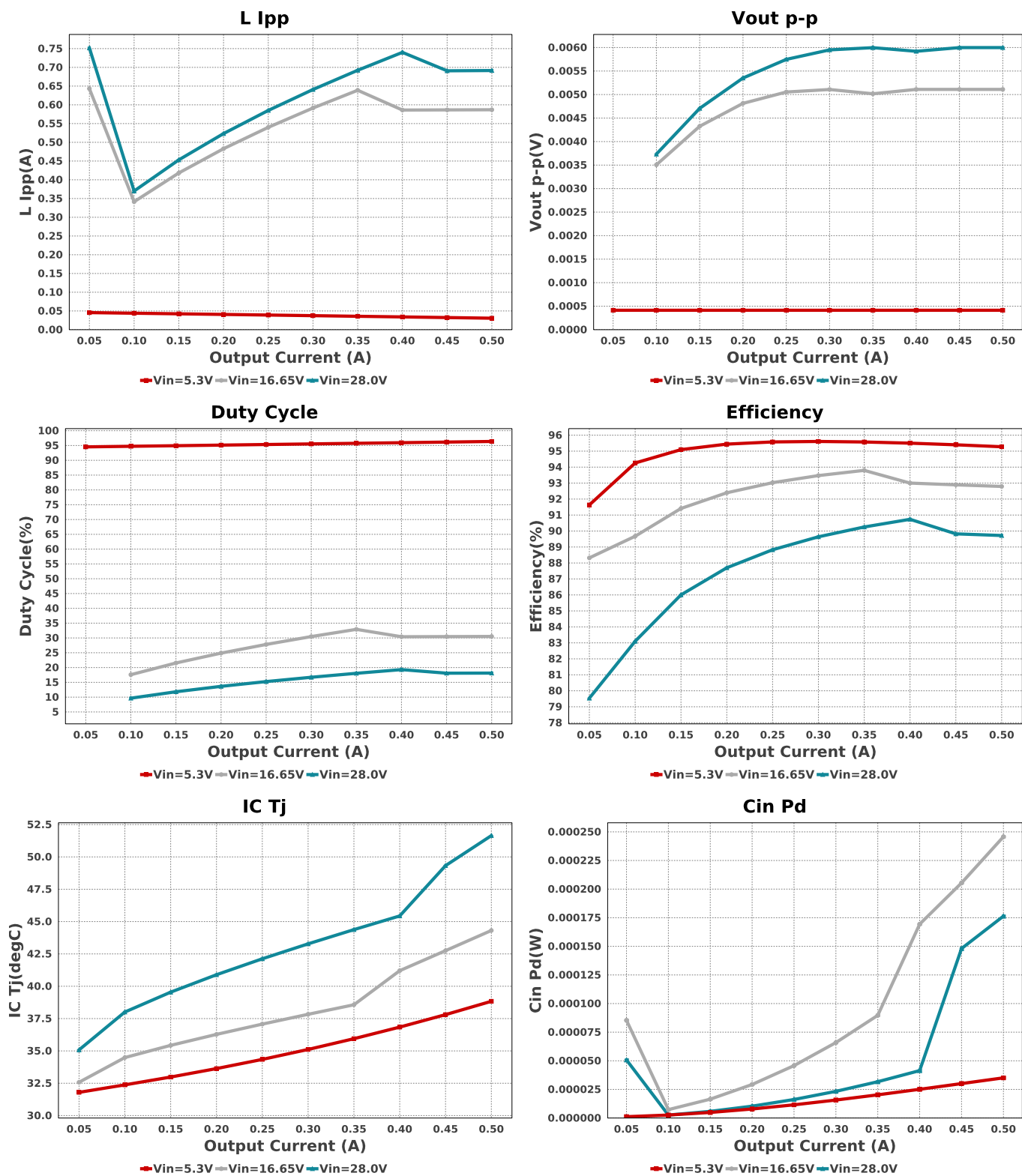
WEBENCH® Design Report

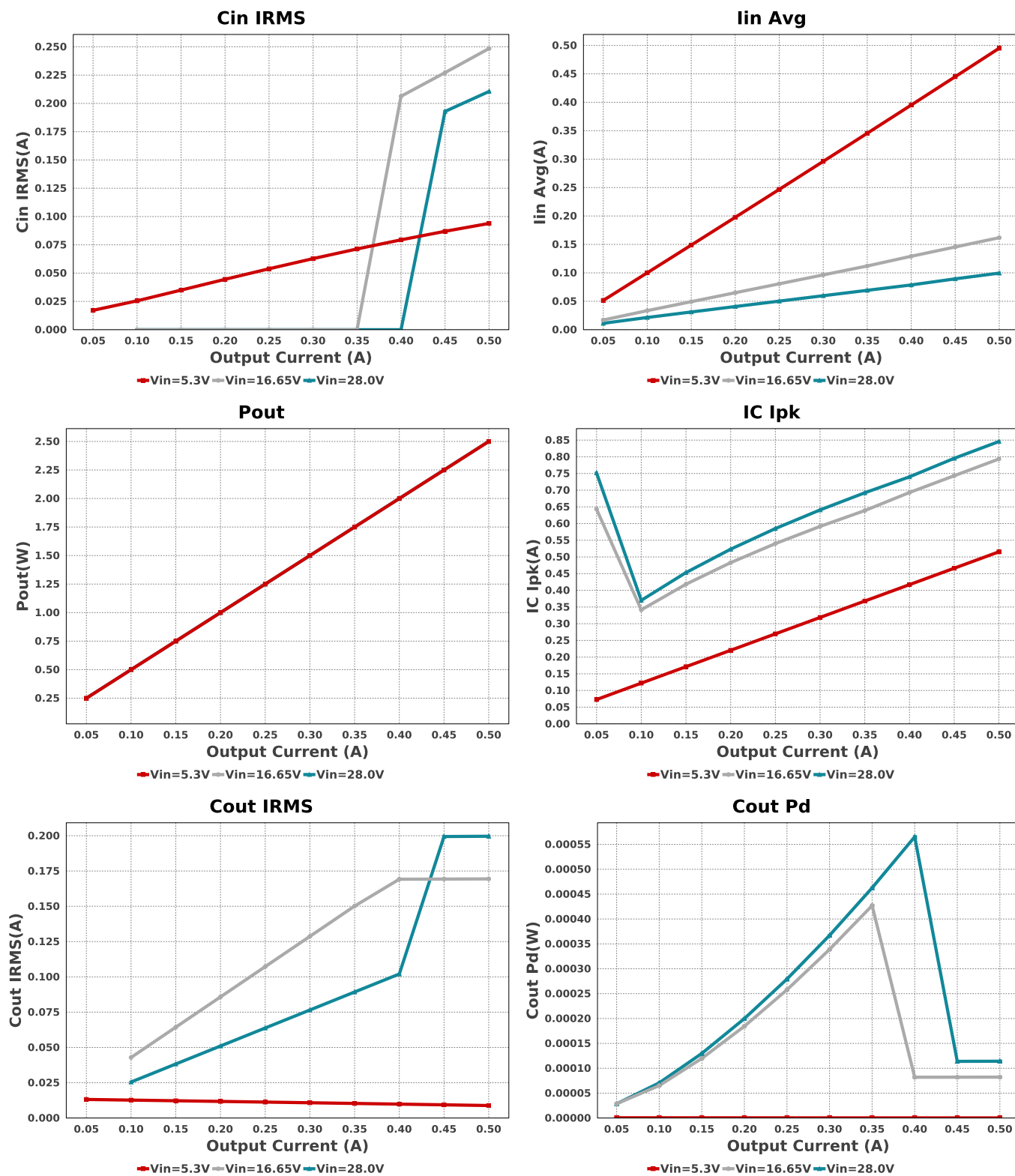
Design : 44 TPS54202HDDCR
TPS54202HDDCR gbg-pcb-2.0

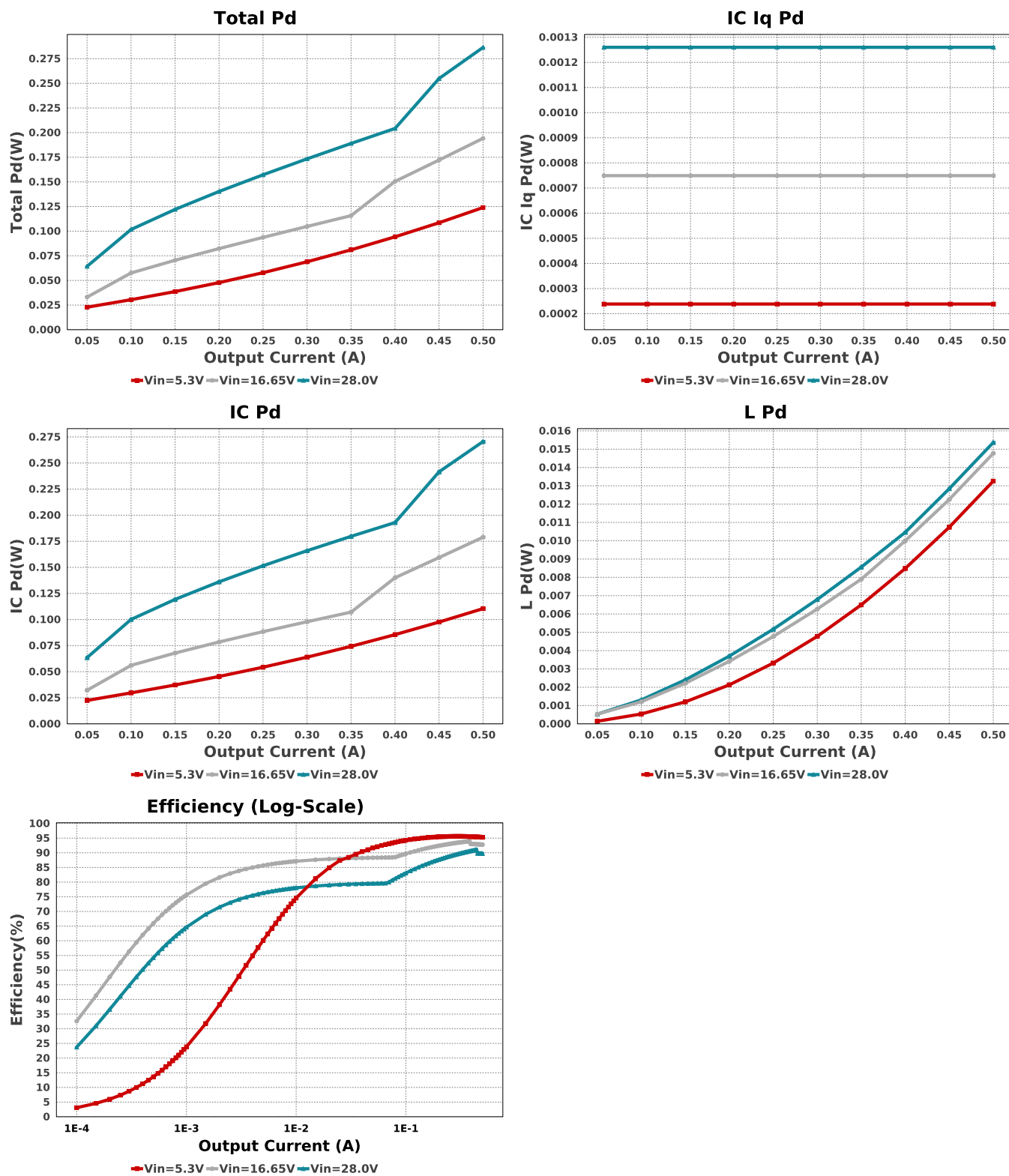


Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbst	TDK	CGA3E2X7R1H104K080AA Series= X7R	Cap= 100.0 nF ESR= 29.6 mOhm VDC= 50.0 V IRMS= 971.99 mA	1	\$0.01	 0603 5 mm²
Cff	Yageo	CC0603JRNPO9BN560 Series= C0G/NP0	Cap= 56.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	 0603 5 mm²
Cin	Taiyo Yuden	MSASU32MSB5106KPNA01 Series= X5R	Cap= 10.0 uF ESR= 3.979 mOhm VDC= 50.0 V IRMS= 3.4821 A	1	\$0.16	 1210 15 mm²
Cout	MuRata	GRM32ER71A476KE15L Series= X7R	Cap= 47.0 uF ESR= 2.864 mOhm VDC= 10.0 V IRMS= 4.8625 A	1	\$0.35	 1210_280 15 mm²
L1	Bourns	SRR6038-120Y	L= 12.0 uH 53.0 mOhm	1	\$0.62	 IND_NPI54C 47 mm²
Ren	Vishay-Dale	CRCW0805590KFKEA Series= CRCW..e3	Res= 590.0 kOhm Power= 125.0 mW Tolerance= 1.0%	1	\$0.01	 0805 7 mm²
Rfbb	Vishay-Dale	CRCW080513K7FKEA Series= CRCW..e3	Res= 13.7 kOhm Power= 125.0 mW Tolerance= 1.0%	1	\$0.01	 0805 7 mm²
Rfbs	Vishay-Dale	CRCW0805100KFKEA Series= CRCW..e3	Res= 100.0 kOhm Power= 125.0 mW Tolerance= 1.0%	1	\$0.01	 0805 7 mm²
U1	Texas Instruments	TPS54202HDDCR	Switcher	1	\$0.40	 DDC0006A_N 10 mm²







Operating Values

#	Name	Value	Category	Description
1.	BOM Count	9		Total Design BOM count
2.	Total BOM	\$1.581		Total BOM Cost
3.	Cin IRMS	210.537 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	176.37 μ W	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	199.636 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	114.14 μ W	Capacitor	Output capacitor power dissipation
7.	IC Ipk	845.78 mA	IC	Peak switch current in IC
8.	IC Iq Pd	1.26 mW	IC	IC Iq Pd
9.	IC Pd	270.58 mW	IC	IC power dissipation
10.	IC Tj	51.646 degC	IC	IC junction temperature
11.	ICThetaJA Effective	80.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance

#	Name	Value	Category	Description
12.	Iin Avg	99.516 mA	IC	Average input current
13.	L Ipp	691.559 mA	Inductor	Peak-to-peak inductor ripple current
14.	L Pd	15.362 mW	Inductor	Inductor power dissipation
15.	Cin Pd	176.37 µW	Power	Input capacitor power dissipation
16.	Cout Pd	114.14 µW	Power	Output capacitor power dissipation
17.	IC Pd	270.58 mW	Power	IC power dissipation
18.	L Pd	15.362 mW	Power	Inductor power dissipation
19.	Total Pd	286.447 mW	Power	Total Power Dissipation
20.	Duty Cycle	18.126 %	System Information	Duty cycle
21.	Efficiency	89.72 %	System Information	Steady state efficiency
22.	FootPrint	147.0 mm ²	System Information	Total Foot Print Area of BOM components
23.	Frequency	500.0 kHz	System Information	Switching frequency
24.	Iout	500.0 mA	System Information	Iout operating point
25.	Mode	CCM	System Information	PWM/PFM Mode
26.	Pout	2.5 W	System Information	Total output power
27.	Vin	28.0 V	System Information	Vin operating point
28.	Vout Actual	4.98 V	System Information	Vout Actual calculated based on selected voltage divider resistors
29.	Vout Tolerance	1.777 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
30.	Vout p-p	5.999 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	500.0 m	Maximum Output Current
VinMax	28.0	Maximum input voltage
VinMin	5.3	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	TPS54202H	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

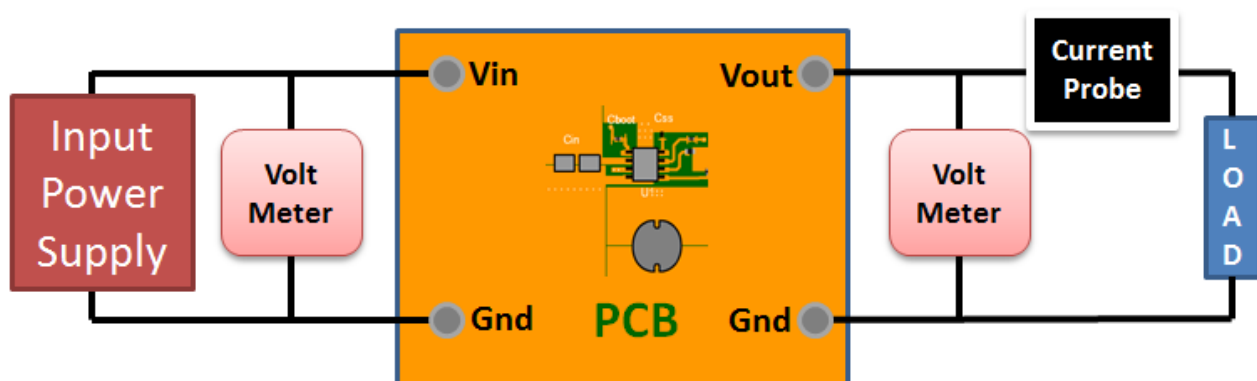
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 5.3V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 36B639548DBC54FC6F0179AABB8E44BC[v1]
2. **TPS54202H** Product Folder : <http://www.ti.com/product/TPS54202H> : contains the data sheet and other resources.

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